

Devdoots - Assignment 02: Python OOPs & Inheritance

Team: Devdoots (The Divine Messengers)

Module: Object-Oriented Programming (Classes, Inheritance, super(), & Method Overriding)

Deadline: Tomorrow before 9:00 PM IST (Next Sync)

1. What & Why (Kya aur Kyun Padhna Hai?)

Object-Oriented Programming (OOPs) Python aur Backend Engineering ki buniyad hai. Inheritance aur Method Overriding seekhna isliye zaroori hai taaki aap scalable, modular, aur standard enterprise code likh sakein bina code duplication ke (DRY Principle: *Don't Repeat Yourself*).

AI/ML Industry Relevance

- **PyTorch & TensorFlow:** Deep Learning mein jab bhi koi custom Neural Network banaya jata hai, toh PyTorch ki base class `torch.nn.Module` ko inherit kiya jata hai.
 - **Hugging Face Transformers:** Custom LLM architectures aur fine-tuning pipelines `PreTrainedModel` class ko inherit karti hain.
 - **Code Reusability:** Production level system mein har baar zero se code nahi likha jata, balki core framework ko extend kiya jata hai.
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2. Assignment Task Specifications

Aapko ek **Smart Role-Based User Access System** design karna hai OOPs ka istemal karke.

Step-by-Step Task Requirements:

1. **Parent Class (BaseUser):**
 - Constructor (`__init__`) mein 2 variables lijiye: username aur email.
 - Ek method banaiye `get_profile()` jo return kare: `f"User: {self.username} | Email: {self.email}"`.
2. **Child Class (AdminUser):**
 - `BaseUser` class ko inherit kare.
 - Constructor mein 3 parameters lijiye: username, email, aur `permissions_level` (e.g., "Full Access").
 - `super().__init__(username, email)` ka use karke username aur email Parent class ko dein.
 - `self.permissions_level = permissions_level` set karein.
 - **Method Overriding:** `get_profile()` method ko override karein taaki woh admin

permissions level bhi dikhaye:

```
f"ADMIN: {self.username} | Email: {self.email} | Permissions: {self.permissions_level}"
```

3. Execution & Testing:

- 1 BaseUser object (`user1`) aur 1 AdminUser object (`admin1`) banayein.
- Dono ke `get_profile()` call karke terminal par print karein.

3. GitHub Submission Rules & Workflow

Direct main branch par code push karna mana hai. Niche diye gaye standard Git workflow ko follow karein:

Step	Command / Action	Description
1. Sync Main Branch	git checkout main git pull origin main	Remote repo se latest code pull karein taaki local main updated rahe.
2. Create Feature Branch	git checkout -b user/yourname-assign2	Apni alag branch banayein (e.g., user/divyansh-assign2).
3. Create Assignment File	Assignment_02_OOPs/yourname_assign2.py	Folder ke andar apni personal file banayein aur code save karein.
4. Commit & Push	git add . git commit -m "[HW-2] completed OOPs assignment 2" git push origin user/yourname-assign2	Proper commit tag ke sath code GitHub par push karein.
5. Raise Pull Request (PR)	GitHub Repo -> Compare & pull request	PR raise karein. Code review ke baad use main mein merge kiya jayega.

Note: Deadline kal raat 9:00 PM IST se pehle tak hai.